De Morgan's Laws & Circuit Analysis

1. De Morgan's Laws

Theory:
De Morgan's Laws provide a mathematical way to break large inversion bars (complements) that cover entire expressions. They are essential for converting NAND and NOR logic into basic AND/OR logic.

- **Law 1 ("Break the line, change the sign"):** The complement of a product is equal to the sum of the complements.

$$1. \quad \overline{x \cdot y} = \overline{x} + \overline{y}$$

- **Law 2 ("Break the line, change the sign"):** The complement of a sum is equal to the product of the complements.

$$2. \quad \overline{x+y} = \overline{x} \cdot \overline{y}$$

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2. Applying De Morgan's Laws (Examples)

Theory:
When dealing with complex expressions, treat large groups of variables as a single temporary variable (like x or y). Once you split the main bar, you can apply De Morgan's law recursively to the smaller sections.

Example A: Complex Expansion

Break the expression down step-by-step. First, define your x and y to split the main addition sign.

Equation:

$$\overline{(A + B\overline{C}) + D(E + \overline{F})}$$

$$\text{Let } x = (A + B\overline{C}) \text{ and } y = D(E + \overline{F})$$

$$\begin{aligned}
&= \overline{(A + B\overline{C})} \cdot \overline{D(E + \overline{F})} && \text{(Applied Law 2)} \\
&= \overline{(A + B\overline{C})} \cdot (\overline{D} + \overline{E + \overline{F}}) && \text{(Applied Law 1 to second term)} \\
&= \overline{(A + B\overline{C})} \cdot (\overline{D} + \overline{E} \cdot \overline{\overline{F}}) && \text{(Applied Law 2 to inner term)} \\
&= \overline{(A + B\overline{C})} \cdot (\overline{D} + \overline{E}F) && \text{(Double inversion cancels out)}
\end{aligned}$$

Example B: Sum of Products Inversion

Equation:

$$\overline{(A+B+C)D}$$

Let $x = (A+B+C)$ and $y = D$

$$\begin{aligned} &= \overline{(A+B+C)} + \overline{D} && \text{(Applied Law 1)} \\ &= \overline{ABC} + \overline{D} && \text{(Applied Law 2 to first term)} \end{aligned}$$

Example C: Grouped Variables

Equation:

$$\overline{ABC + DEF}$$

Let $x = ABC$ and $y = DEF$

$$\begin{aligned} &= \overline{ABC} \cdot \overline{DEF} && \text{(Applied Law 2)} \\ &= (\overline{A} + \overline{B} + \overline{C}) \cdot (\overline{D} + \overline{E} + \overline{F}) && \text{(Applied Law 1)} \end{aligned}$$

Practice Problem 1

Theory: Treat the three main chunks ($A\overline{B}$, $\overline{C}D$, and EF) as individual terms first. Break the main sum into products, and then apply De Morgan's Law again to break the individual products into sums.

Equation:

$$\overline{A\overline{B} + \overline{C}D + EF}$$

$$\begin{aligned} &= \overline{(A\overline{B})} \cdot \overline{(\overline{C}D)} \cdot \overline{(EF)} && \text{(Applied Law 2: Break the main addition signs)} \\ &= (\overline{A} + \overline{\overline{B}}) \cdot (\overline{\overline{C}} + \overline{D}) \cdot (\overline{E} + \overline{F}) && \text{(Applied Law 1 to each product term)} \\ &= (\overline{A} + B)(C + \overline{D})(\overline{E} + \overline{F}) && \text{(Simplified double inversions)} \end{aligned}$$

Practice Problem 2

Theory: This one looks intimidating but is actually the easiest of the three. It is a great test of realizing that a double bar over an entire expression completely cancels out.

Equation:

$$\overline{\overline{(A+B)} + \overline{C}}$$

$$\begin{aligned} &= \overline{\overline{(A+B)}} \cdot \overline{\overline{C}} && \text{(Applied Law 2: Break the main addition line)} \\ &= (A+B) \cdot C && \text{(Simplified double inversions)} \end{aligned}$$

Practice Problem 3

Theory: Similar to the first problem, break the main sum into three distinct products first. Be careful to keep parentheses around the CD term when you break its bar so you don't accidentally ruin the order of operations.

Equation:

$$
\begin{aligned}
&\overline{\overline{A} + B + CD} \\
&= \overline{\overline{A}} \cdot \overline{B} \cdot \overline{CD} && \text{(Applied Law 2: Break the main addition signs)} \\
&= A \cdot \overline{B} \cdot \overline{CD} && \text{(Simplified double inversion on A)} \\
&= A\overline{B}(\overline{C} + \overline{D}) && \text{(Applied Law 1 to break the CD product)}
\end{aligned}
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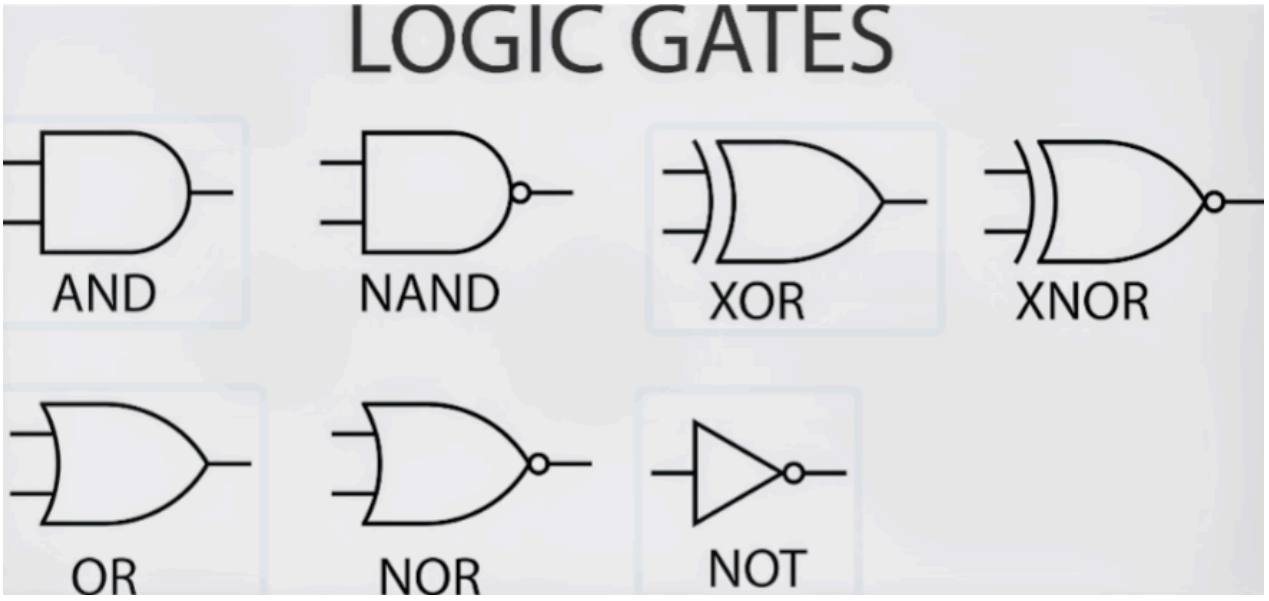
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3. Boolean Analysis of Logic Gates

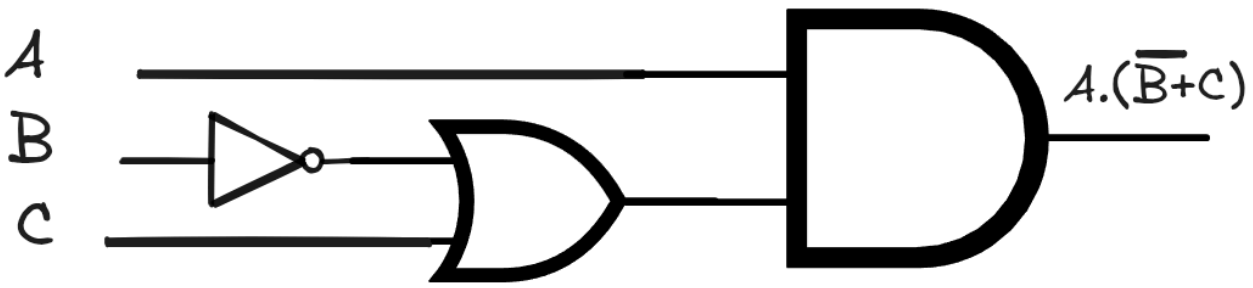
Theory:

To build a physical circuit from a Boolean expression , we map the mathematical operators to their corresponding hardware logic gates.

- Addition ($+$) $\rightarrow$ **OR Gate**
- Multiplication ($\cdot$) $\rightarrow$ **AND Gate**
- Complement ($-$) $\rightarrow$ **NOT Gate (Inverter)**
- XOR ($\oplus$) $\rightarrow$ **Exclusive-OR Gate**



Design: $y = A \cdot (\overline{B} + C)$


A
B
C
A.(B+C)